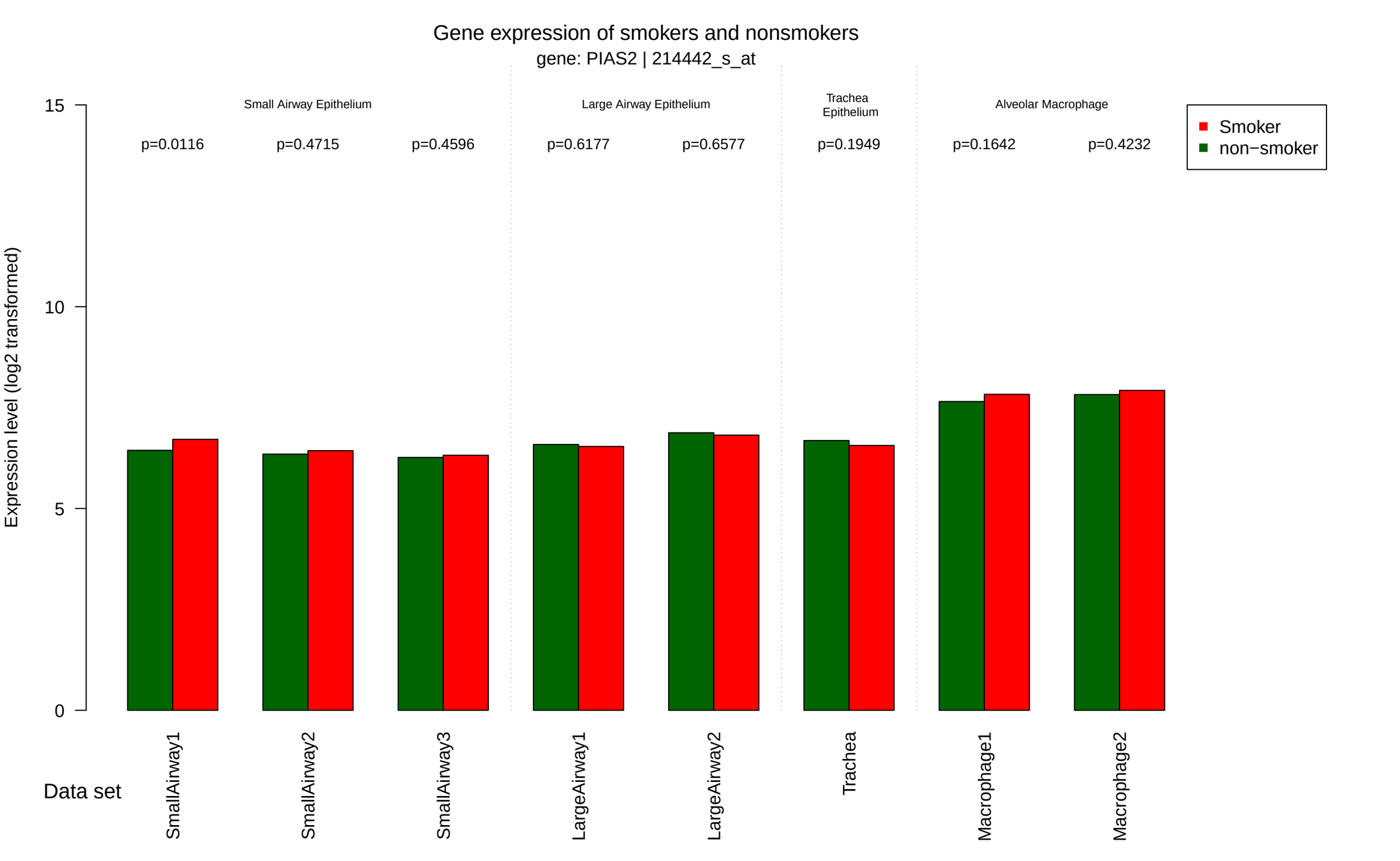
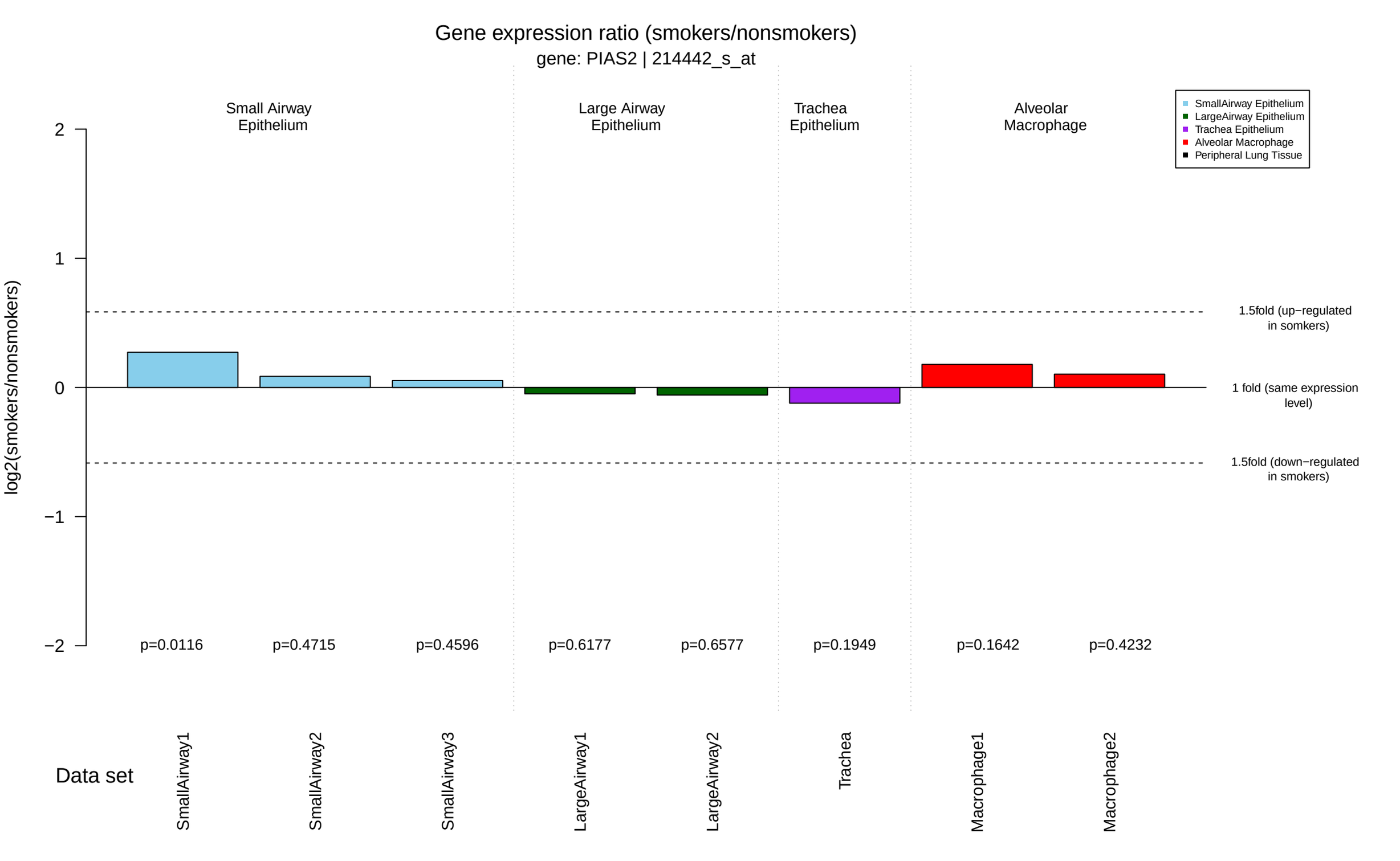
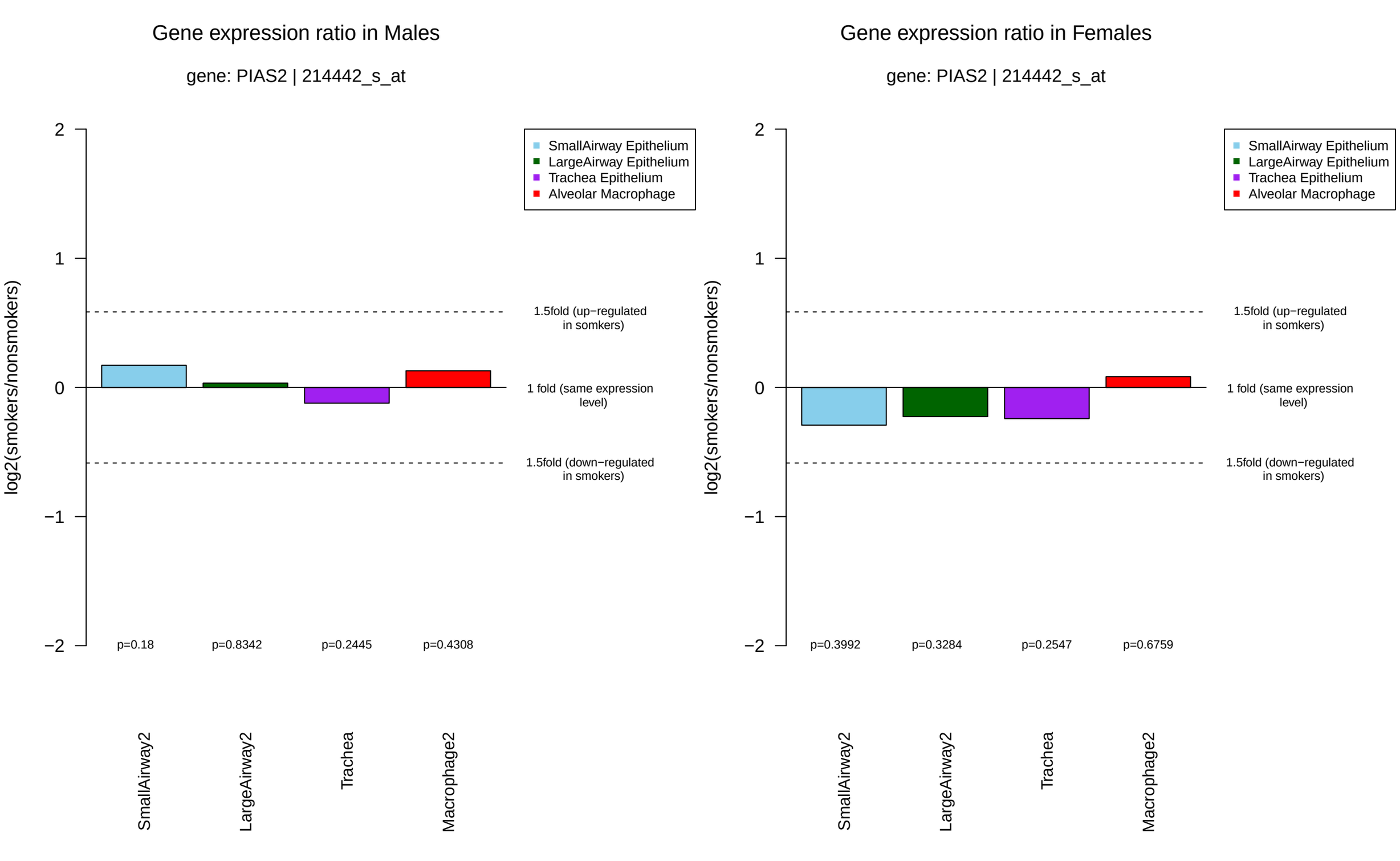
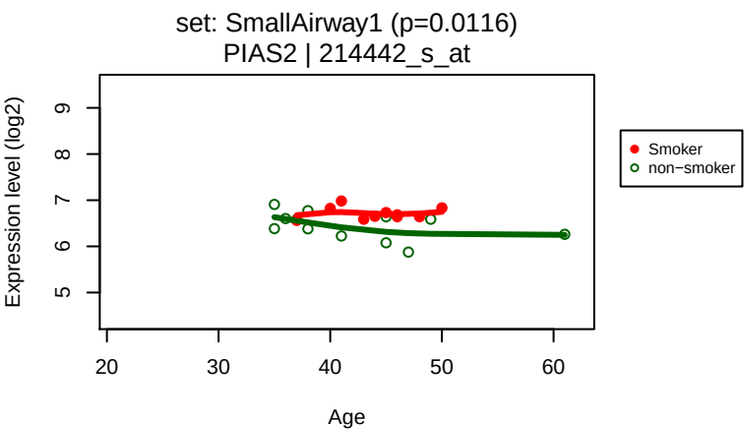
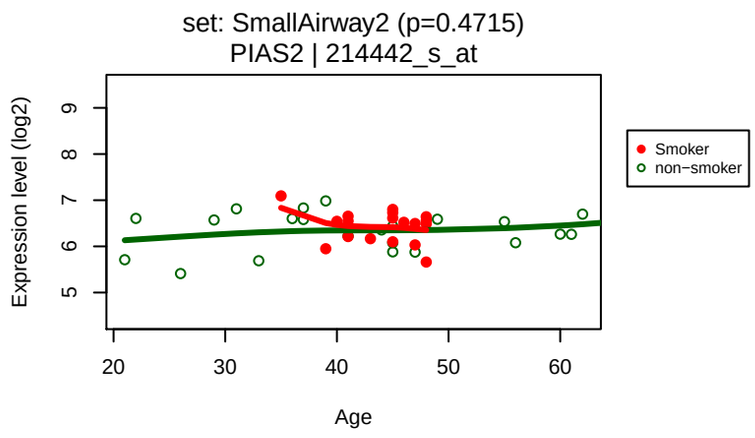
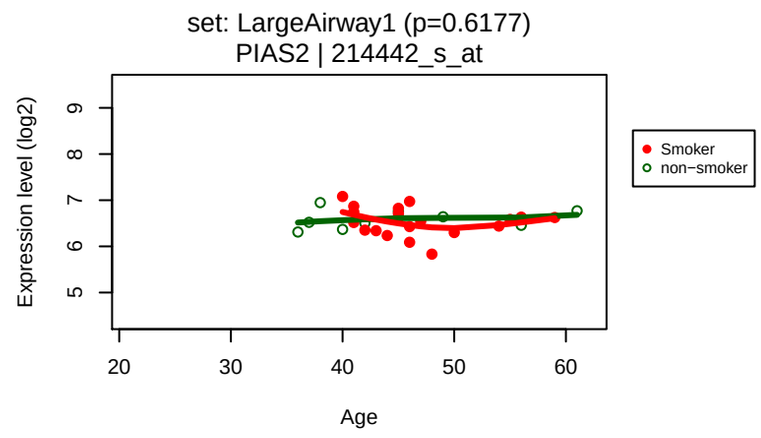

Gene expression of smokers and nonsmokers
gene: PIAS2 | 214442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0116
p=0.4715
p=0.4596
p=0.6177
p=0.6577
p=0.1949
p=0.1642
p=0.4232
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

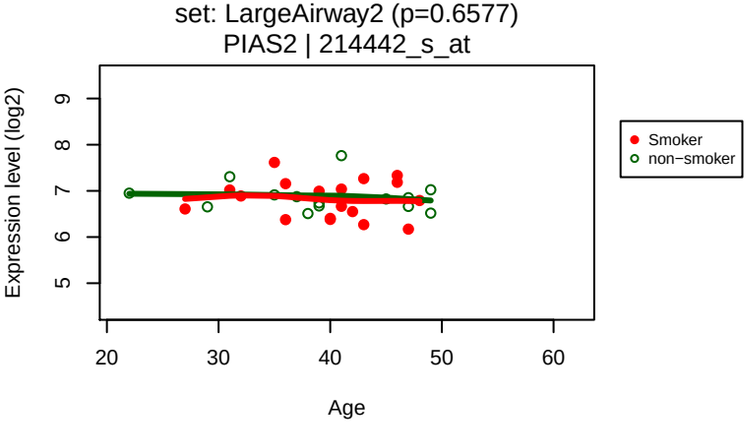

Gene expression ratio (smokers/nonsmokers)
gene: PIAS2 | 214442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0116
p=0.4715
p=0.4596
p=0.6177
p=0.6577
p=0.1949
p=0.1642
p=0.4232
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

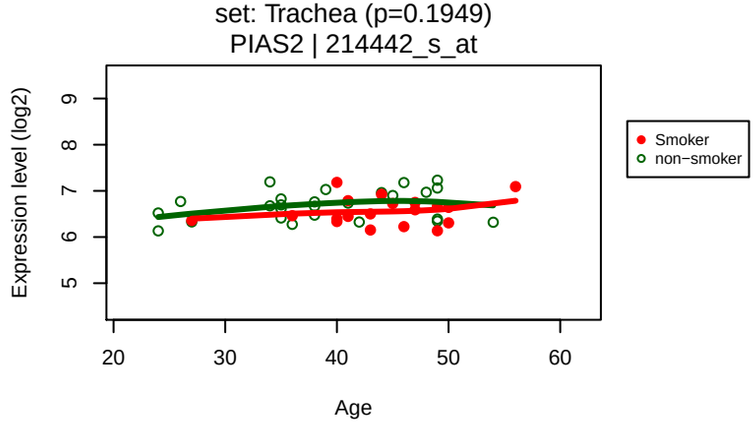

Gene expression ratio in Males
gene: PIAS2 | 214442_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.18
p=0.8342
p=0.2445
p=0.4308
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS2 | 214442_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3992
p=0.3284
p=0.2547
p=0.6759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

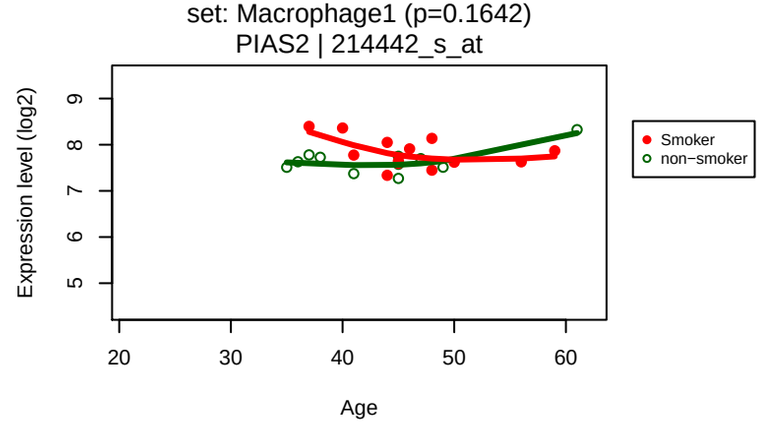

set: SmallAirway1 (p=0.0116)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker

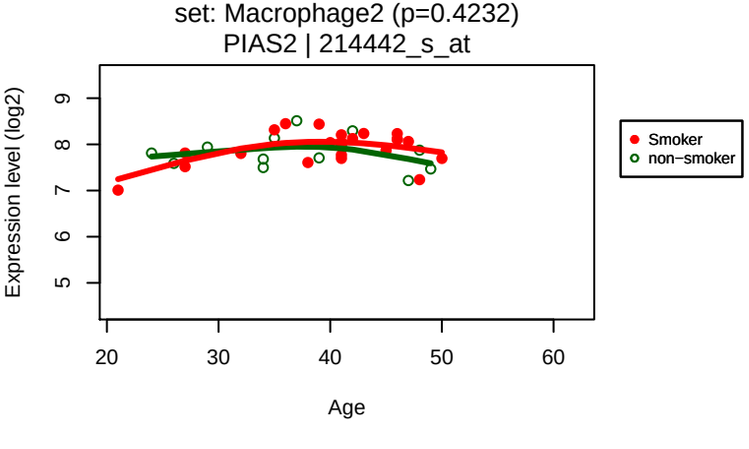

set: SmallAirway2 (p=0.4715)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker

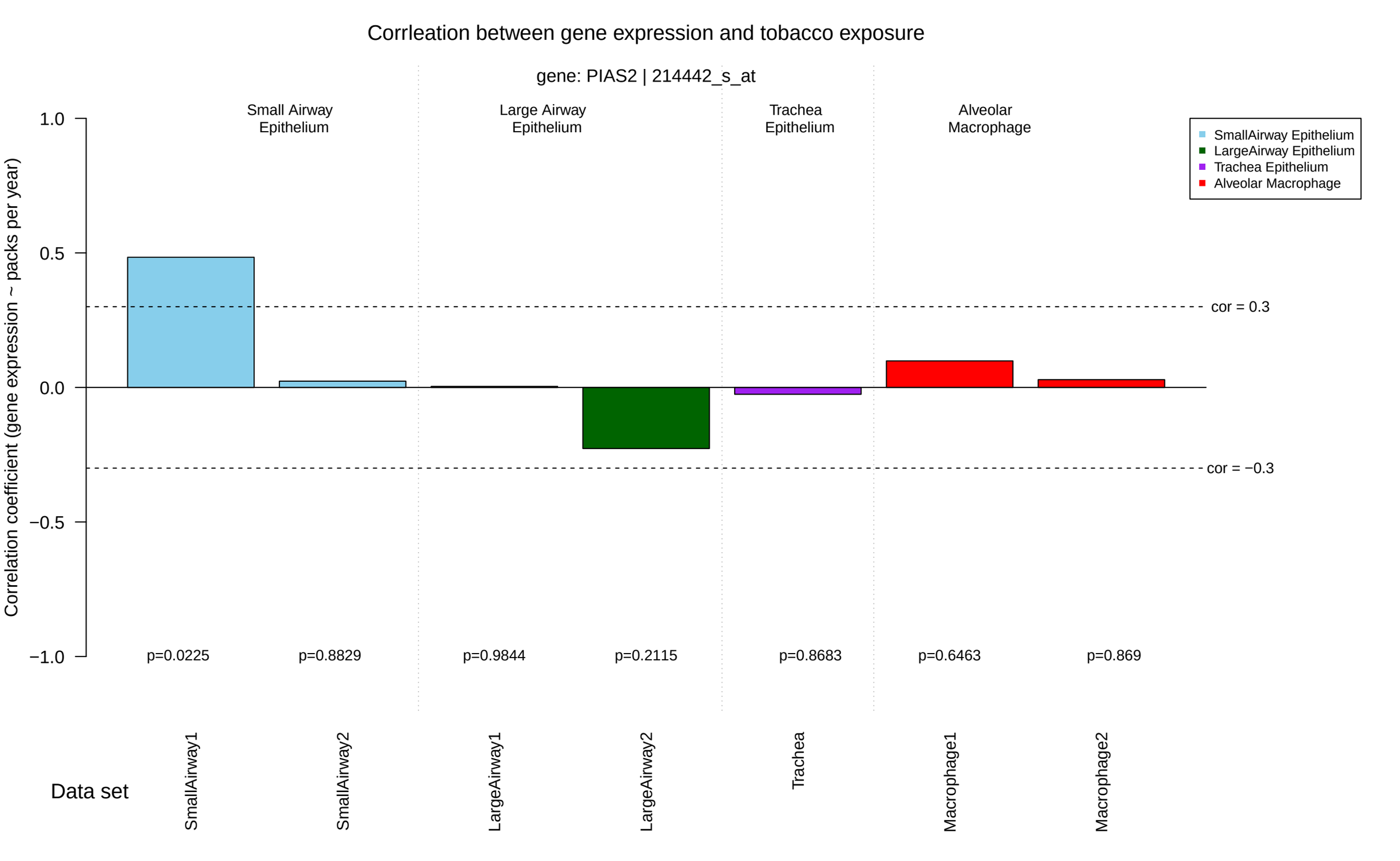

set: LargeAirway1 (p=0.6177)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6577)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1949)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1642)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4232)
PIAS2 | 214442_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS2 | 214442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0225
p=0.8829
p=0.9844
p=0.2115
p=0.8683
p=0.6463
p=0.869
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2